

TAHA SHAHEEN

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SUMMARY

Ph.D. student in Computer Science at Arizona State University working on human-robot interaction and collaboration. Background in teleoperated robots, human-subjects studies, preference-based reinforcement learning, and engineering low-cost robots for use in autism therapy. Seeking internships for Summer 2026.

Availability: Seeking a full-time, 12-week internship for Summer 2026.

EDUCATION

Ph.D. Computer Science Arizona State University, USA <i>Jumpstart Research Grant, Teaching Excellence Award (2025)</i>	Expected May 2028 4.00/4.00 GPA
M.S. Social Informatics Kyoto University, Japan <i>MEXT Research Scholarship (Japanese Gov.), Distinguished Master's Thesis Award</i>	April 2023 3.92/4.00 GPA
B.E. Electronic Engineering NED University of Engineering and Technology, Pakistan <i>Academic Rank 5 out of 148</i>	November 2017 3.70/4.00 GPA

RESEARCH AND ENGINEERING EXPERIENCE

Graduate Student Researcher <i>Cooperative Robotic Systems Lab, Arizona State University, USA</i>	August 2023 – Present
- Engineered a full-stack data collection platform (Python, Flask, JS) for a large-scale HRI study, successfully collecting over 2,550 unique trajectories from 102+ participants in a custom grid-world environment. - Designed and executing a 16-condition study, examining the impact of perturbed user beliefs on feedback quality in reward learning for robots, in collaboration with ASU's most-cited researcher, Dr. Stephen G. West. - Accelerated the development cycle by an estimated 30-40% through AI-assisted rapid prototyping and deployed an automated pipeline for data collection and processing, minimizing manual intervention.	
Graduate Student Researcher <i>Human-Robot Interaction Lab, Kyoto University, Japan</i>	October 2020 – August 2023
- Devised a novel role-playing study to explore adversarial behaviors yielding key insights for future robotic ethics frameworks where 12 participants used both a physical robot and a crowded simulation to act as malicious operators and victims. - Identified and cataloged 15 unique "low-moral" actions (e.g., privacy violation, harassment), and developed a framework of technical and social countermeasures to prevent them. - Authored the resulting taxonomy and prevention framework, culminating in a first-author publication in the journal <i>ACM Transactions on Human-Robot Interaction</i>.	
Research Assistant <i>National Center for Artificial Intelligence, NED University, Pakistan</i>	November 2018 – September 2020
- Engineered 2 socially assistive robots from concept to deployment for a target cost under \$200 USD, developing the full software/hardware stack in Unity (C#), Android (Java), and Arduino (C++). - Drove 3 major robot iterations by incorporating insights from a 3-month field study and targeted feedback from 15 key stakeholders (therapists and parents). - Mentored 12 undergraduate students across 3 capstone teams on robotics projects for autism therapy, leading all 3 teams to successfully develop and validate novel therapeutic robot prototypes.	

TECHNICAL SKILLS

Programming Languages: Python, C++, Java, C#, JavaScript, R, Kotlin, MIPS Assembly
Libraries & Frameworks: PyTorch, ROS 2, Pandas, NumPy, Scikit-learn, Flask, OpenCV, Matplotlib
Software & Tools: Git, Docker, Unity, Fusion 360, Android Studio, Google Dialogflow
Hardware & Robotics: Universal Robots UR5, Raspberry Pi, Arduino

NOTABLE PROJECTS

LLM as a Model Optimizer in Model Predictive Control 2025

- **Proved** that an LLM can function as a real-time model optimizer for an MPC system, successfully adapting the controller parameters online to stabilize a dynamic inverted pendulum simulation.
- **Engineered** a physical testbed for hardware validation by designing and **3D-printing** a custom pendulum and integrating it with a **UR5 robotic arm**.

Active Shadowing (ASD) for Implicit HRI Communication 2025

- **Contributed to** the development of Active Shadowing, a novel system using Augmented Reality shadows to enhance robot collaboration without sacrificing motion optimality.
- **Developed** the core robot control software, precisely **choreographing** a UR5 arm's movements to seamlessly integrate with a predictive Augmented Reality shadow projected via a HoloLens.
- **Co-designed** the 23-participant user study that validated the system, proving via t-test ($p < 0.05$) that the AR shadow improved human-robot safety scores by over **50%** compared to explicit communication.

Office Reception Robot 2020

- **Engineered** the robot's perception and interaction systems, integrating a Python-based computer vision module for facial recognition and Google Dialogflow for natural language understanding (NLU).
- **Built** the full-stack architecture, including a Python back-end and 3 custom Android front-ends, to successfully automate **3 core office tasks** (guest registration, package delivery, and meeting scheduling).

Assistive Social Robot (Capstone Project) 2017

- **Co-led** a team of 4 to design, build, and deploy a social robot from the ground up, developing the full stack using Unity (C#), Android (Java), and Arduino (C++).
- The project's success and technical novelty directly led to securing a **\$26,000** grant from Pakistan's Higher Education Commission (TDF-02145), which funded its continuation as a formal research position.

PUBLICATIONS

Andrew Boateng, Prakhar Bhartiya, **Taha Shaheen**, Yu Zhang. 2025. "Active Shadowing (ASD): Manipulating Perception of Robotic Behaviors via Implicit Virtual Communication." arXiv preprint arXiv: arXiv:2407.01468

Winston Smith, Andrew Boateng, **Taha Shaheen**, and Yu Zhang. 2025. "Assigning Multi-Robot Tasks to Multitasking Robots." arXiv preprint arXiv:2506.15032.

Taha Shaheen, Dražen Bršćić, and Takayuki Kanda. 2024. Investigation of Low-Moral Actions by Malicious Anonymous Operators of Avatar Robots. *ACM Transactions on Human-Robot Interaction*. doi.org/10.1145/3696466.

LEADERSHIP & VOLUNTEER EXPERIENCE

Oral History Apprentice & Field Officer 2016 – 2019
1947 Partition Archive

Led a team of 3 volunteers and personally interviewed and filmed 83 witnesses of the 1947 Partition of British India, with work featured on BBC Radio and in Stanford University's library collection.